

Chapter 29: Unemployment

If the economy is growing, why might millions of people still be unemployed?



Learning Objectives

29.1 Calculate the labor force statistics using economic data.

29.2 Distinguish between employment categories

29.3 Differentiate among types of unemployment—frictional, structural, and cyclical—and analyze their causes.

29.2 Evaluate policies and institutions (e.g., unemployment insurance, unions, and collective bargaining) and their effects on unemployment.

29.4 Explain why the economy experiences a natural rate of unemployment even in the long run.

Key Points

- The unemployment rate is the percentage of those who would like to work but don't have jobs. The Bureau of Labor Statistics calculates this statistic monthly based on a survey of thousands of households.
- The unemployment rate is an imperfect measure of joblessness. Some people who call themselves unemployed may actually not want to work, and some who would like to work have left the labor force after an unsuccessful search and therefore are not counted as unemployed.
- In the U.S. economy, most people who become unemployed find work within a short period of time. Nonetheless, most unemployment observed at any given time is attributable to the few people who are unemployed for long periods.

Key Points

- One reason for unemployment is the time it takes for workers to search for jobs that best suit their tastes and skills. Unemployment insurance, a government policy designed to protect workers' incomes, increases frictional unemployment by reducing the search effort of the unemployed.
- A second reason the economy always has some unemployment is minimum-wage laws. By raising the wage of unskilled and inexperienced workers above the equilibrium level, these laws increase the quantity of labor supplied and reduce the quantity demanded. The resulting surplus of labor represents unemployment.
- A third reason for unemployment is the market power of unions. When unions push the wages in unionized industries above the equilibrium level, they create a surplus of labor.
- A fourth reason for unemployment is suggested by the theory of efficiency wages. According to this theory, firms find it profitable to pay wages above the equilibrium level. Higher wages can improve worker health, reduce worker turnover, raise worker quality, increase worker effort, and boost worker morale

Labor Force Statistics

- Produced by Bureau of Labor Statistics (BLS), in the U.S. Dept. of Labor
- Based on regular survey of 60,000 households
- Based on “adult population” (16 years or older)
 - Civilian noninstitutional population
 - Doesn't include persons found in prisons, mental institutions, or the military

Labor Force Statistics

- BLS divides population into 3 groups:
 - **Employed**: paid employees, self-employed, and unpaid workers in a family business
 - **Unemployed**: people not working who have looked for work during previous 4 weeks
 - **Not in the labor force**: everyone else
- The **labor force** is the total # of workers, including the employed and unemployed.

Labor Force Statistics

Unemployment rate (“u-rate”):

% of the labor force that is unemployed

$$\text{u-rate} = 100 \times \frac{\text{\# of unemployed}}{\text{labor force}}$$

Labor force participation rate:

% of the adult population that is in the labor force

$$\text{labor force participation rate} = 100 \times \frac{\text{labor force}}{\text{adult population}}$$

Labor Force Statistics Example

- Compute the labor force, unemployment rate, adult population, and labor force participation rate using this data:

<u>Adult population of the U.S.</u> <u>by group</u> , August, 2025 (seasonally adjusted)	
# of employed	163,394 (thousands)
# of unemployed	7,384 (thousands)
not in labor force	103,223 (thousands)

Note, the link above updates monthly

Labor Force Statistics Example: Answers

$$\begin{aligned}\text{Labor force} &= \text{employed} + \text{unemployed} \\ &= 163,394 + 7,384 \\ &= 170,778 \text{ thousands}\end{aligned}$$

$$\begin{aligned}\text{U-rate} &= 100 \times (\text{unemployed})/(\text{labor force}) \\ &= 100 \times (7,384/163,394) \\ &= 4.3\%\end{aligned}$$

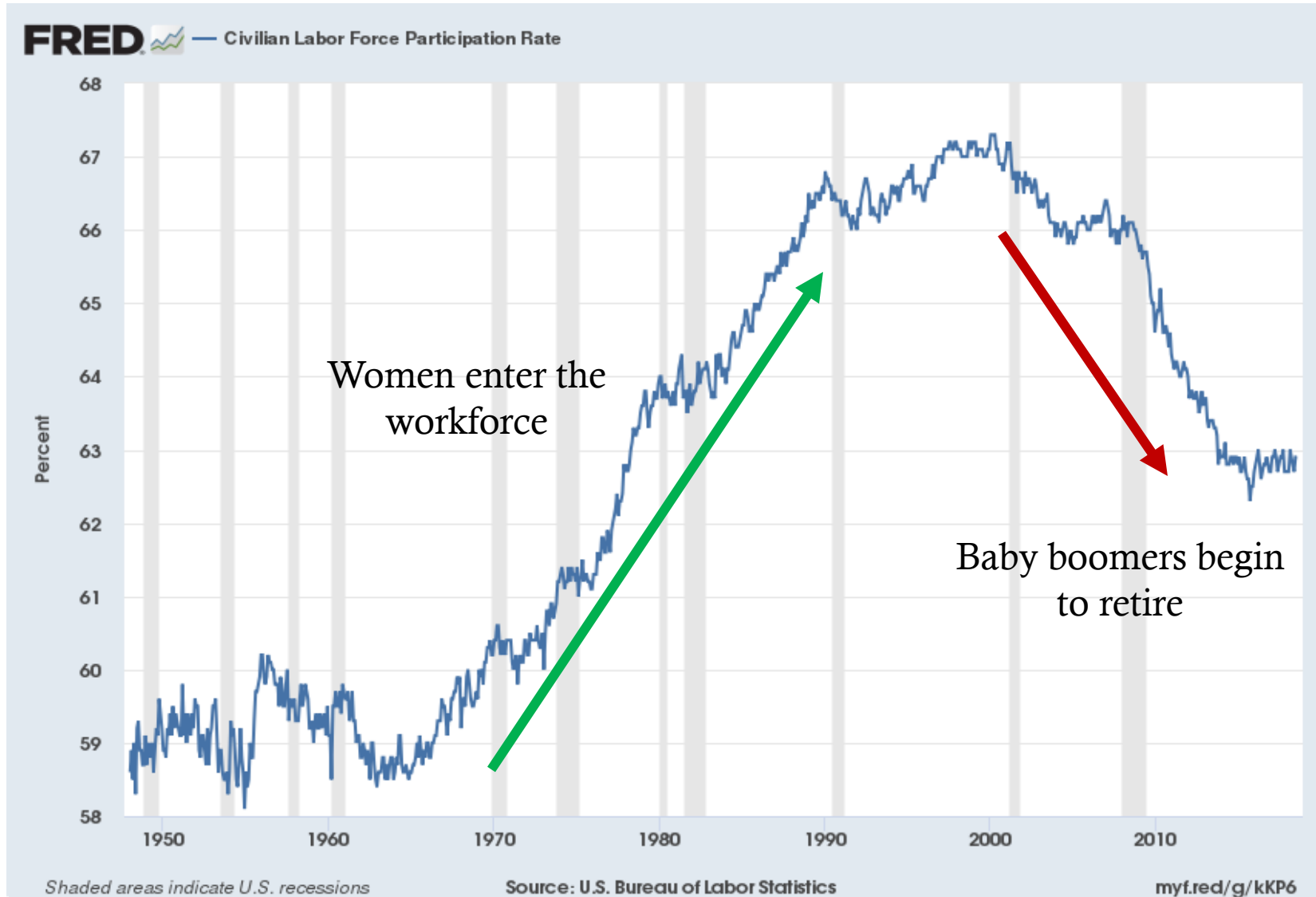
Labor Force Statistics Example: Answers

$$\begin{aligned}\text{Population} &= \text{labor force} + \text{not in labor force} \\ &= 170,778 + 103,223 \\ &= 274,001 \text{ thousands}\end{aligned}$$

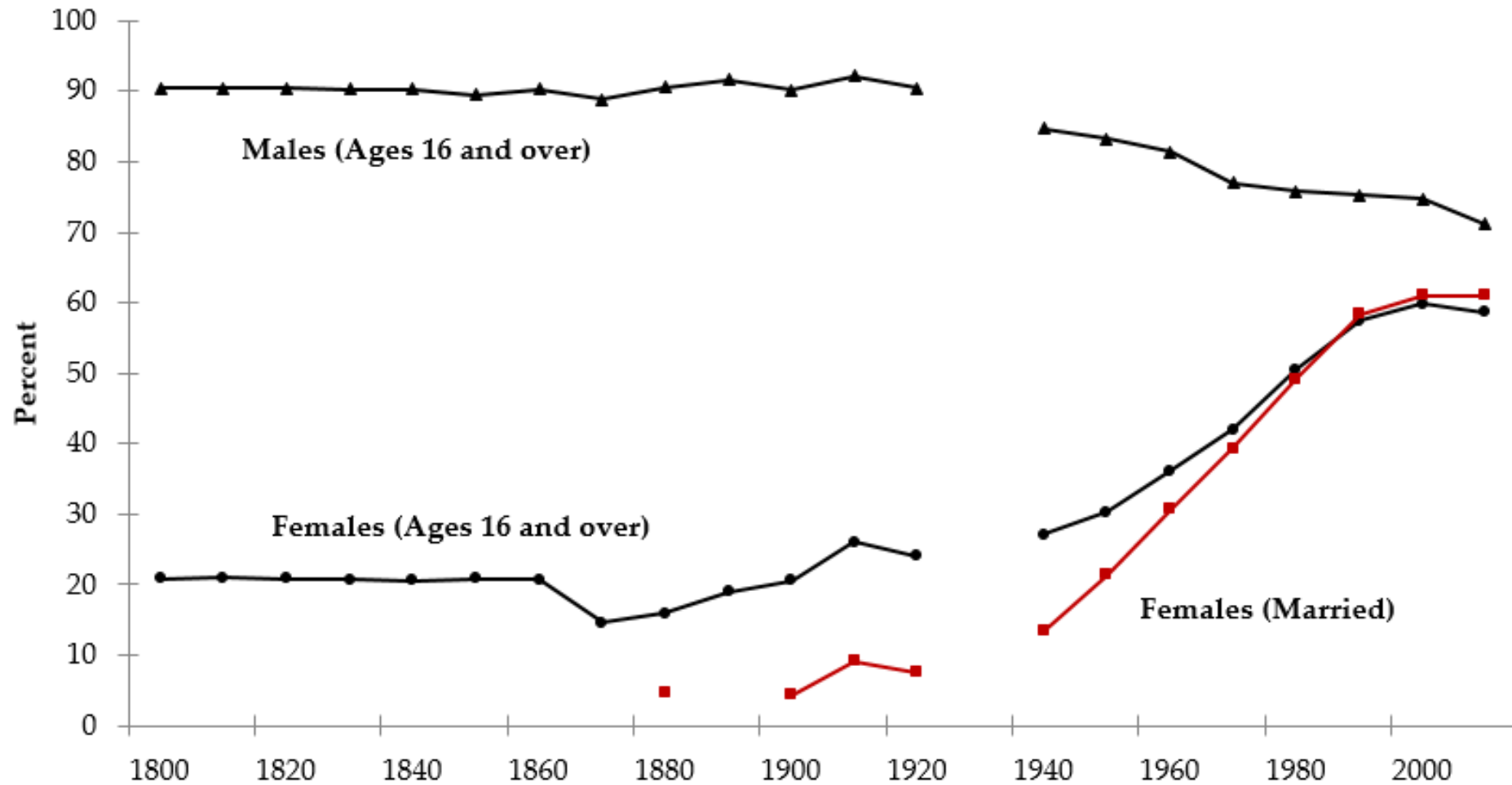
Labor Force Participation Rate

$$\begin{aligned}&= 100 \times (\text{labor force})/(\text{population}) \\ &= 100 \times (170,778/274,001) \\ &= 62.3\%\end{aligned}$$

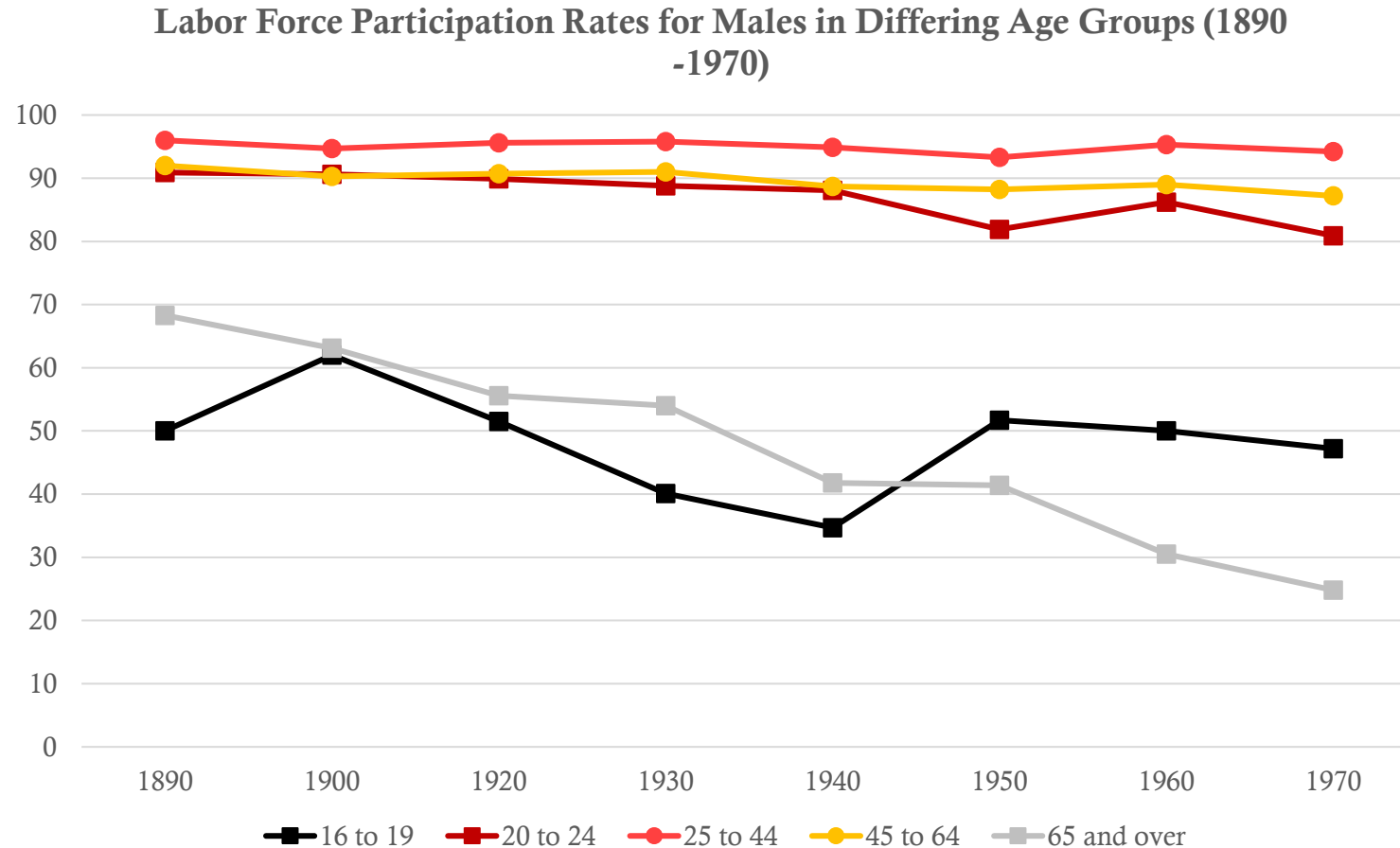
Labor Statistics Data



Labor Force Participation Rates by Sex

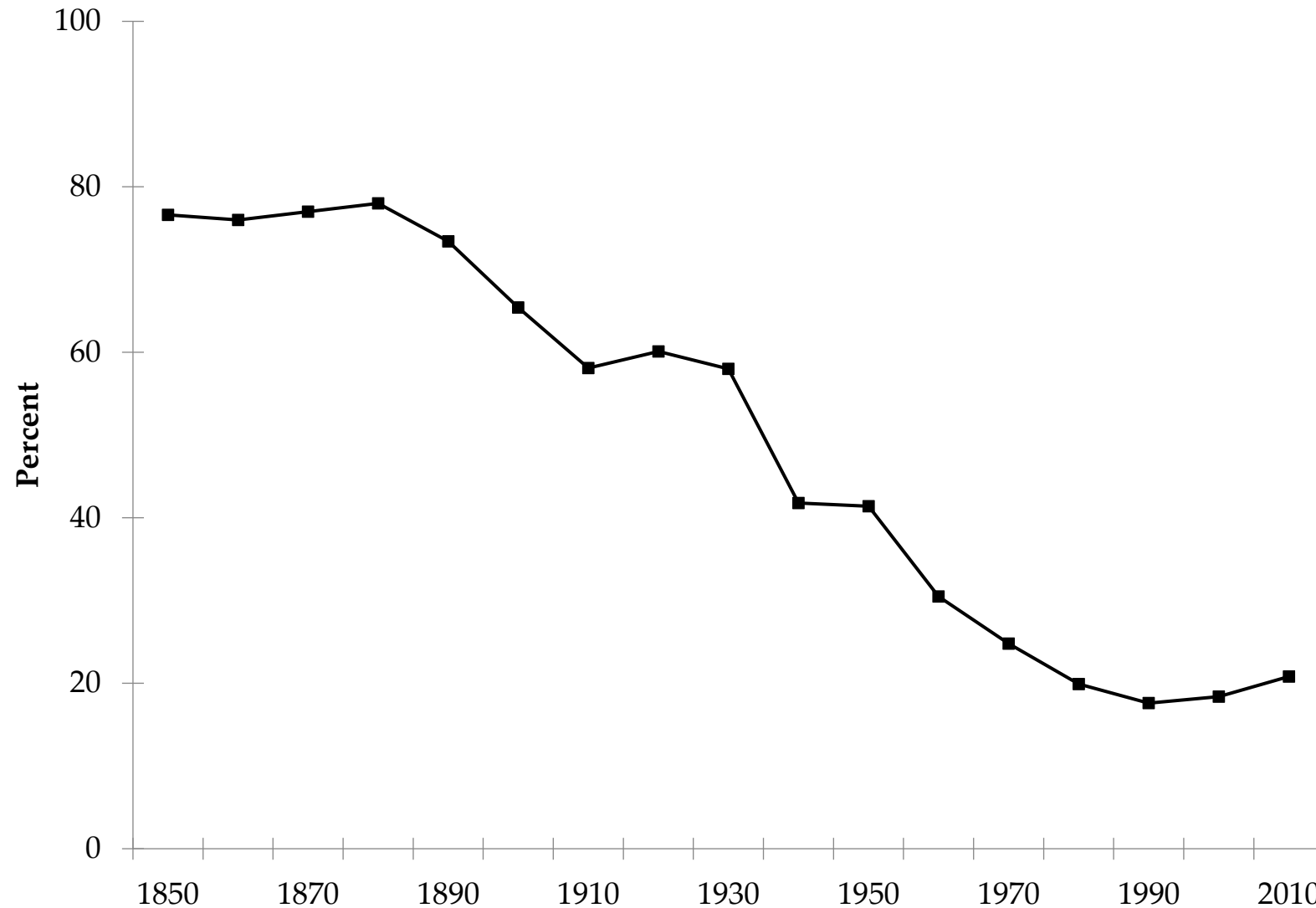


Labor Force Participation Rates for Males by Age



Source: United States. Bureau of the Census. (1975). *Historical statistics of the United States, colonial times to 1970* (No. 93). US Department of Commerce, Bureau of the Census, Series D 31-35.

Labor Force Participation Rates for Males, 1850-2010: Ages 65 and Older



Activity: Limitations of the Unemployment Rate

- In each of the following, what happens to the unemployment rate? Does the unemployment rate give an accurate impression of what's happening in the labor market?
 1. Sue lost her job and begins looking for a new one.
 2. Jon, a steelworker who has been out of work since his mill closed last year, becomes discouraged and gives up looking for work.
 3. Sam, the sole earner in his family of 5, just lost his \$80,000 job as a research scientist. Immediately, he takes a part-time job at McDonald's until he can find another job in his field.

Activity: Limitations

1. Sue lost her job and begins looking for a new one.

- **u-rate rises**

- A rising unemployment rate gives the impression that the labor market is worsening, and it is.
- If the above situation occurs in a manner such that job losers outweigh job entrants, then we will see a rising unemployment rate

Activity: Limitations

2. Jon has been out of work since last year, becomes discouraged, stops looking for work.

Discouraged workers

- would like to work but have given up looking for jobs
- classified as “not in the labor force” rather than “unemployed”
- **U-rate falls** because Jon is no longer counted as unemployed.
- A falling u-rate gives the impression that the labor market is improving, but it is not.

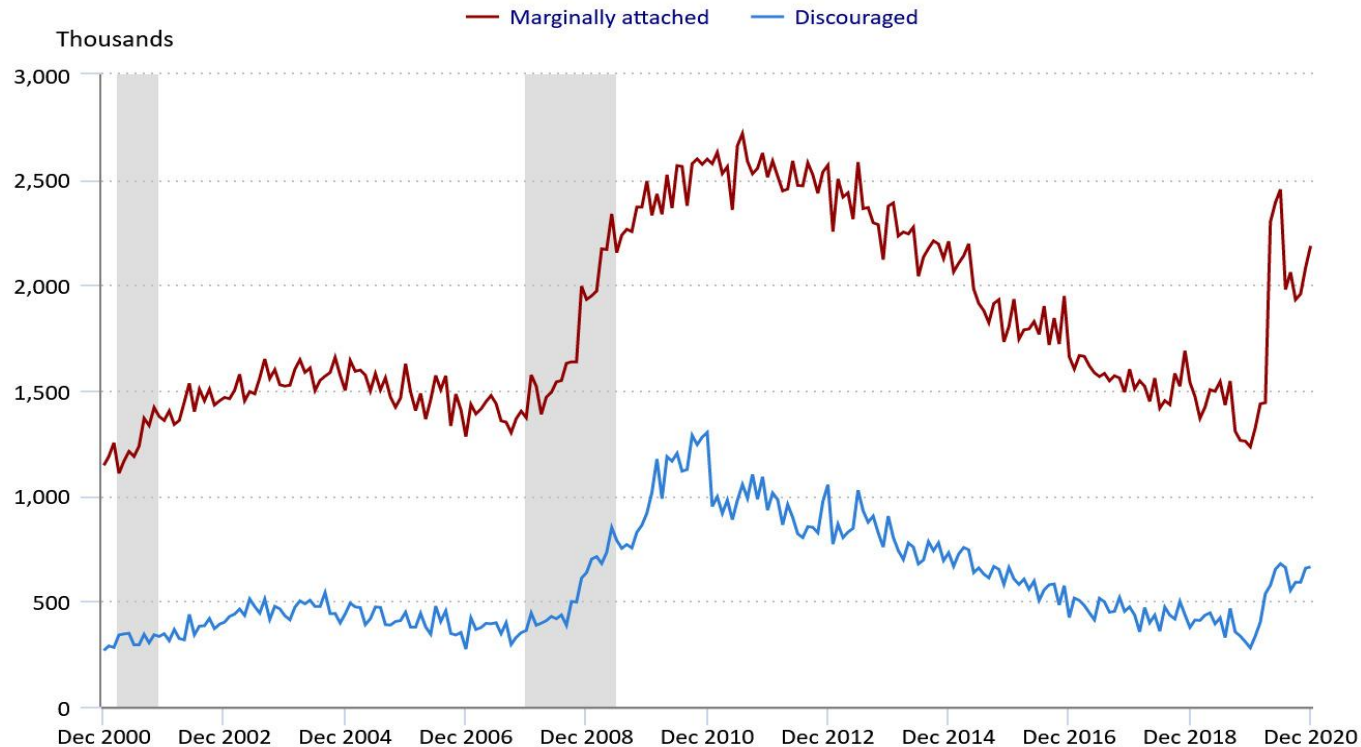
Limitations: Answers

3. Sam lost his \$80,000 job and takes a part-time job at McDonald's until he finds a better one.
- **Unemployment rate unchanged** because a person is “employed” whether they work full or part time.
 - Things are worse, but the unemployment rate fails to show it.

Discourage Worker History

Persons not in the labor force, selected indicators, seasonally adjusted

Click and drag within the chart to zoom in on time periods



Hover over chart to view data.

Note: Shaded area represents recession, as determined by the National Bureau of Economic Research.

Source: U.S. Bureau of Labor Statistics.



From the BLS

Discouraged workers are a subset of persons marginally attached to the labor force. The marginally attached are those persons not in the labor force who want and are available for work, and who have looked for a job sometime in the prior 12 months but were not counted as unemployed because they had not searched for work in the 4 weeks preceding the survey. Among the marginally attached, discouraged workers were not currently looking for work specifically because they believed no jobs were available for them or there were none for which they would qualify.

What Does the Unemployment Rate Really Measure?

- The unemployment rate is not a perfect indicator of joblessness or the health of the labor market:
 - It excludes discouraged workers and marginally attached workers.
 - It does not distinguish between full-time and part-time work, or people working part time because full-time jobs not available.
 - Some people misreport their work status in the BLS survey.
- Despite these issues, the unemployment rate is still a very useful barometer of the labor market & economy.

Calculate Labor Force Statistics

- Suppose that 1 million unemployed workers become discouraged.

<u>Adult population of the U.S.</u> <u>by group,</u> March 2021 (seasonally adjusted)	
# of employed	150.493 million
# of unemployed	9.905 million
not in labor force	100.606 million

Labor Force Statistics Answers

$$\begin{aligned}\text{Labor force} &= \text{employed} + \text{unemployed} \\ &= 150.493 + 8.905 \\ &= 159.398 \text{ million}\end{aligned}$$

$$\begin{aligned}\text{Unemployment rate} &= 100 \times (\text{unemployed})/(\text{labor force}) \\ &= 100 \times (8.905 / 159.398) \\ &= 5.6\%\end{aligned}$$

Employment Situation Summary

Table A-1. Employment status of the civilian population by sex and age

HOUSEHOLD DATA

Table A-1. Employment status of the civilian population by sex and age

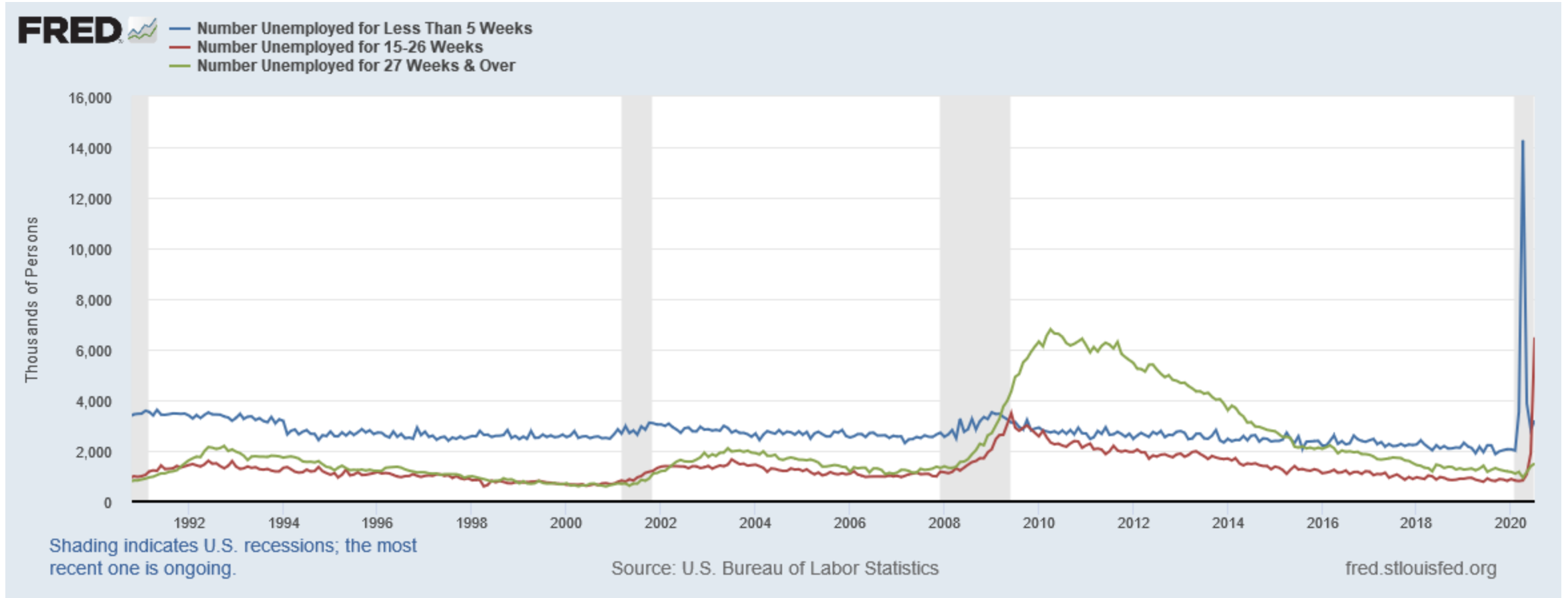
[Numbers in thousands]

Employment status, sex, and age	Not seasonally adjusted			Seasonally adjusted ⁽¹⁾					
	Mar. 2020	Feb. 2021	Mar. 2021	Mar. 2020	Nov. 2020	Dec. 2020	Jan. 2021	Feb. 2021	Mar. 2021
TOTAL									
Civilian noninstitutional population	259,758	260,918	261,003	259,758	261,085	261,230	260,851	260,918	261,003
Civilian labor force	162,537	160,008	160,397	162,721	160,536	160,567	160,161	160,211	160,558
Participation rate	62.6	61.3	61.5	62.6	61.5	61.5	61.4	61.4	61.5
Employed	155,167	149,522	150,493	155,536	149,809	149,830	150,031	150,239	150,848
Employment-population ratio	59.7	57.3	57.7	59.9	57.4	57.4	57.5	57.6	57.8
Unemployed	7,370	10,486	9,905	7,185	10,728	10,736	10,130	9,972	9,710
Unemployment rate	4.5	6.6	6.2	4.4	6.7	6.7	6.3	6.2	6.0
Not in labor force	97,221	100,910	100,606	97,037	100,548	100,663	100,690	100,708	100,445
Persons who currently want a job	5,215	6,902	6,576	5,490	7,127	7,331	6,957	6,933	6,850

The Duration of Unemployment

- Most spells of unemployment are short:
 - Typically, 1/3 of the unemployed have been unemployed under 5 weeks, 2/3 have been unemployed under 14 weeks.
 - Only 20% have been unemployed over 6 months.
- Yet, most observed unemployment is long term.
 - The small group of long-term unemployed persons has fairly little turnover, so it accounts for most of the unemployment observed over time.
- Knowing these facts helps policymakers design better policies to help the unemployed.

The Duration of Unemployment



Cyclical vs the Natural Rate of Unemployment

There's always some unemployment, though the unemployment rate fluctuates from year to year.

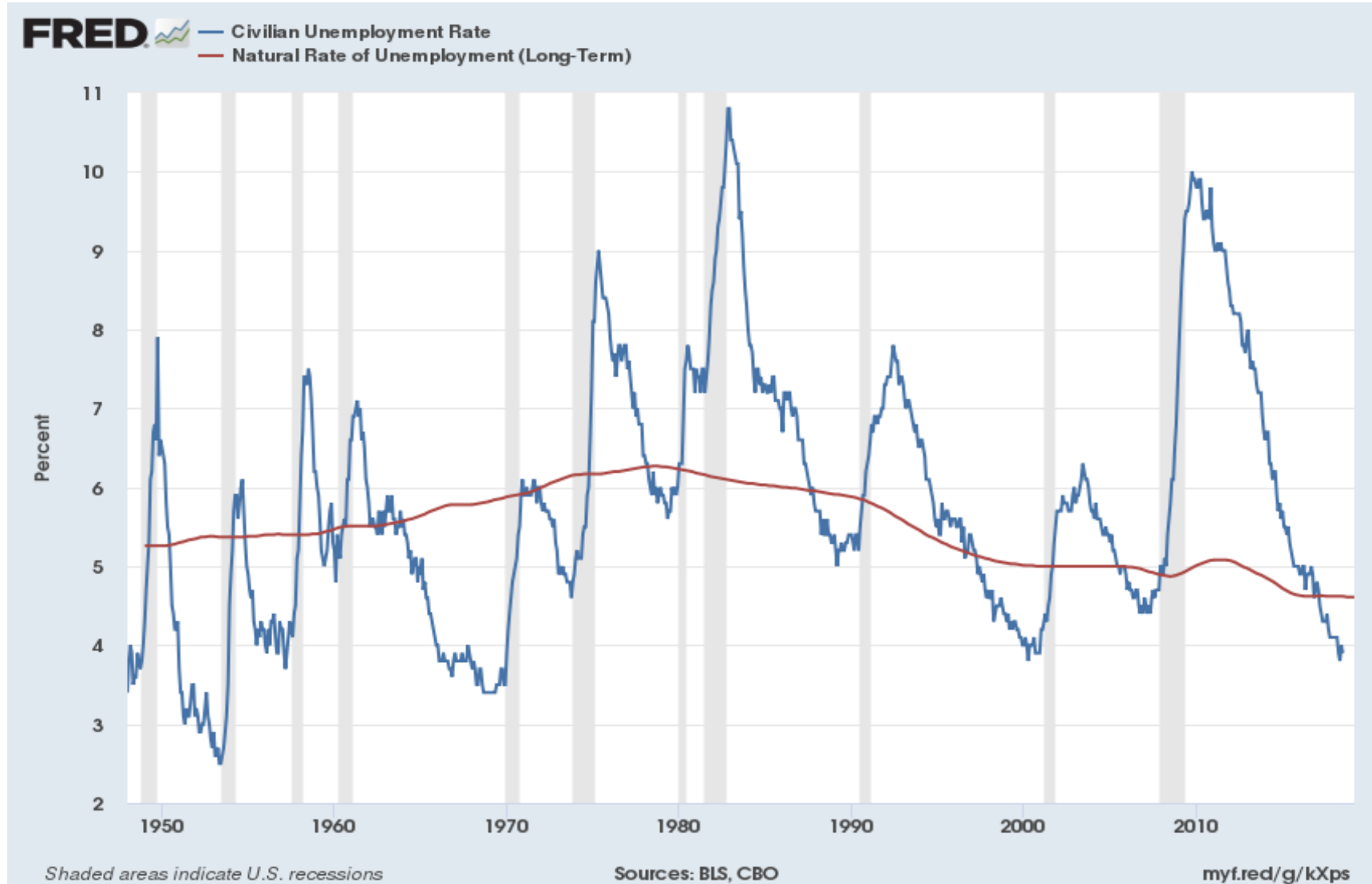
Natural rate of unemployment

- the normal rate of unemployment around which the actual unemployment rate fluctuates

Cyclical unemployment

- the deviation of unemployment from its natural rate
- associated with business cycles, which we'll study in later chapters

U.S. Unemployment, 1948–2019



Explaining the Natural Rate: An Overview

- Even when the economy is doing well, there is always some unemployment, including:
 - **Frictional unemployment**
 - occurs when workers spend time searching for the jobs that best suit their skills and tastes
 - short-term for most workers
 - **Structural unemployment**
 - occurs when there are fewer jobs than workers
 - usually longer-term

Job Search

- Workers have different tastes & skills, and jobs have different requirements.
- **Job search** is the process of matching workers with appropriate jobs.
- **Sectoral shifts** are changes in the composition of demand across industries or regions of the country.
- Such shifts displace some workers, who must search for new jobs appropriate for their skills & tastes.
- The economy is always changing, so some frictional unemployment is inevitable.

Public Policy and Job Search

- *Government employment agencies*
provide information about job vacancies to speed up the matching of workers with jobs.
- *Public training programs*
aim to equip workers displaced from declining industries with the skills needed in growing industries.

Unemployment Insurance

- **Unemployment insurance** (UI):
a government program that partially protects workers' incomes when they become unemployed
- UI increases frictional unemployment.
To see why, recall one of the Ten Principles of Economics:

People respond to incentives.

- UI benefits end when a worker takes a job, so workers have less incentive to search or take jobs while eligible to receive benefits.

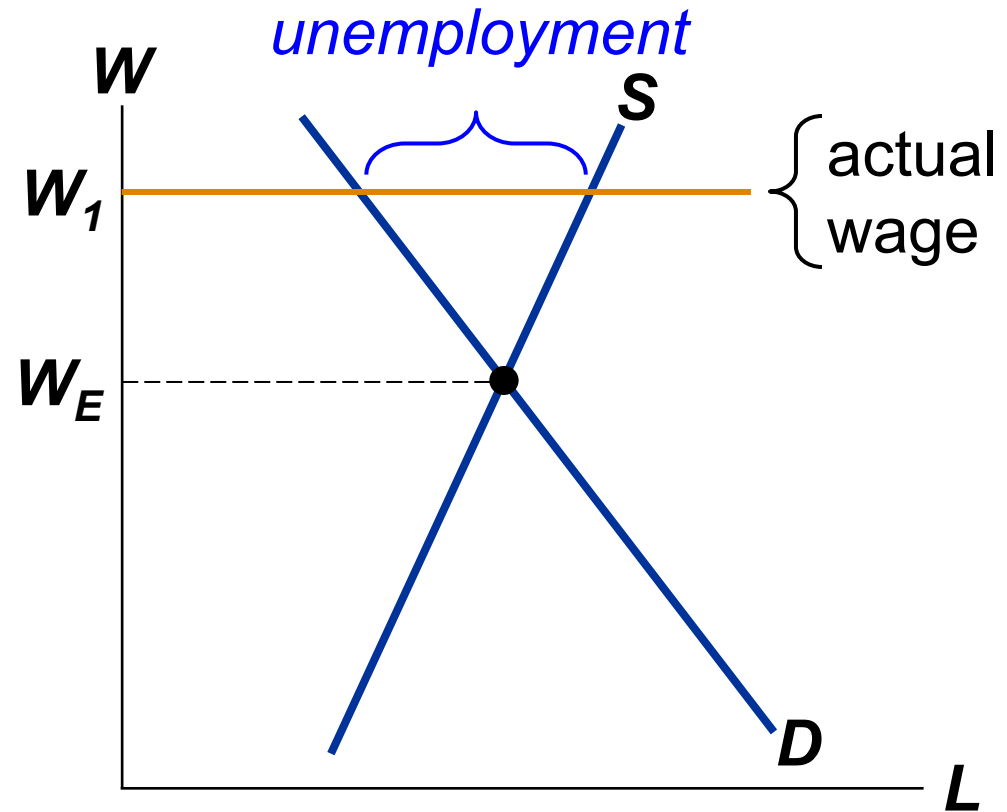
Unemployment Insurance

Benefits of UI:

- Reduces uncertainty over incomes
- Gives the unemployed more time to search, resulting in better job matches and thus higher productivity

Explaining Structural Unemployment

- Structural unemployment occurs when there's not enough jobs to go around.
- Occurs when wage is kept above equilibrium.
- There are three reasons for this...



Three Reasons: Minimum-Wage Laws

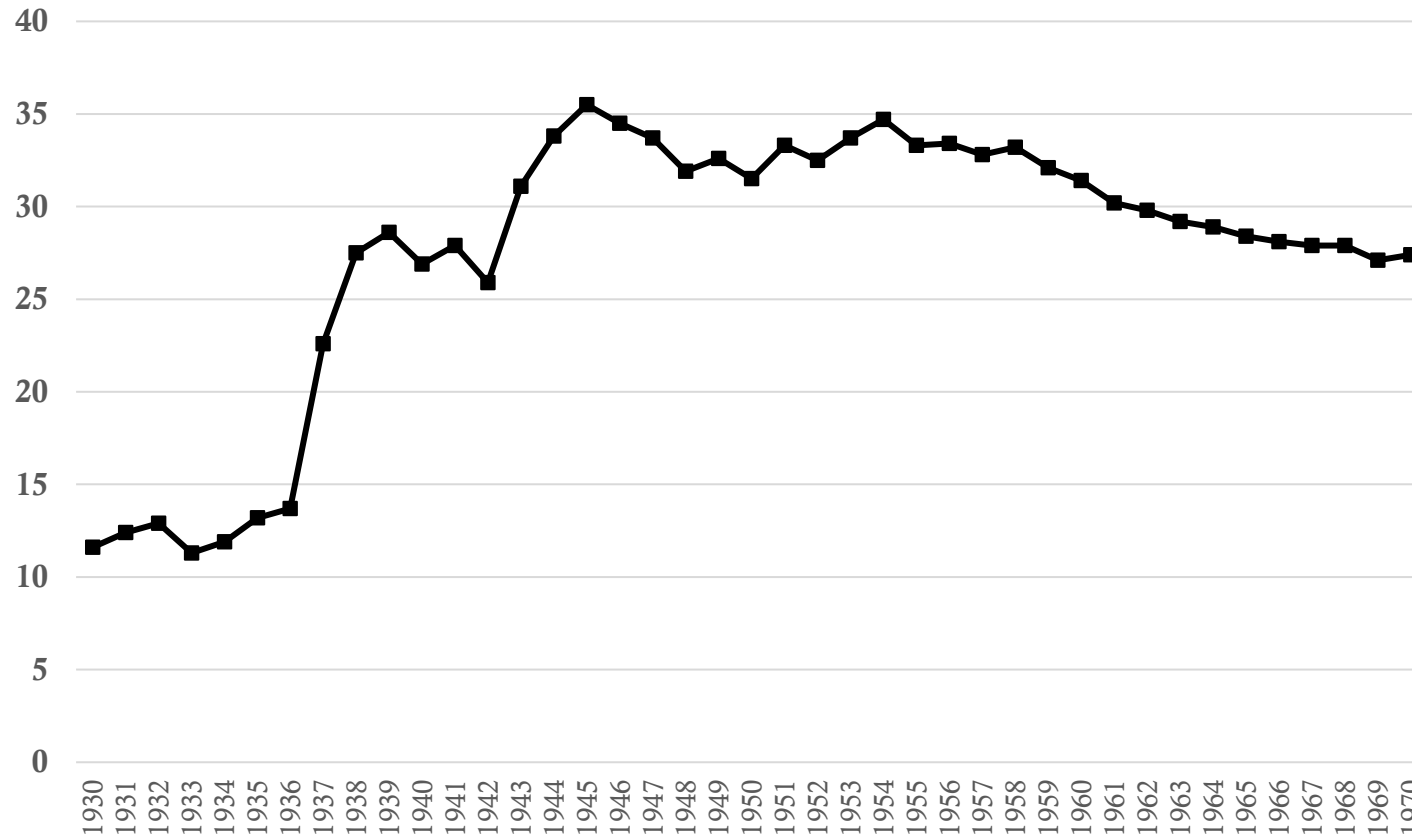
- The minimum wage may exceed the equilibrium wage for the least skilled or experienced workers, causing structural unemployment.
- But this group is a small part of the labor force, so the minimum wage can't explain most unemployment.

Three Reasons: Unions

- **Union**: a worker association that bargains with employers over wages, benefits, and working conditions
 - **Collective Bargaining**: the process by which unions and firms agree on the terms of employment
 - Wagner Act (1935): gave legality to unions
- Unions exert their market power to negotiate higher wages for workers.
- The typical union worker earns 20% higher wages and gets more benefits than a nonunion worker for the same type of work.
 - “Among full-time wage and salary workers, union members had median usual weekly earnings of \$1,144 in 2020, while those who were not union members had median weekly earnings of \$958. In addition to coverage by a collective bargaining agreement, these earnings differences reflect a variety of influences, including variations in the distributions of union members and nonunion employees by occupation, industry, age, firm size, or geographic region, as well as pandemic-related labor market effects in 2020.”
 - ([BLS Union Summary](#))

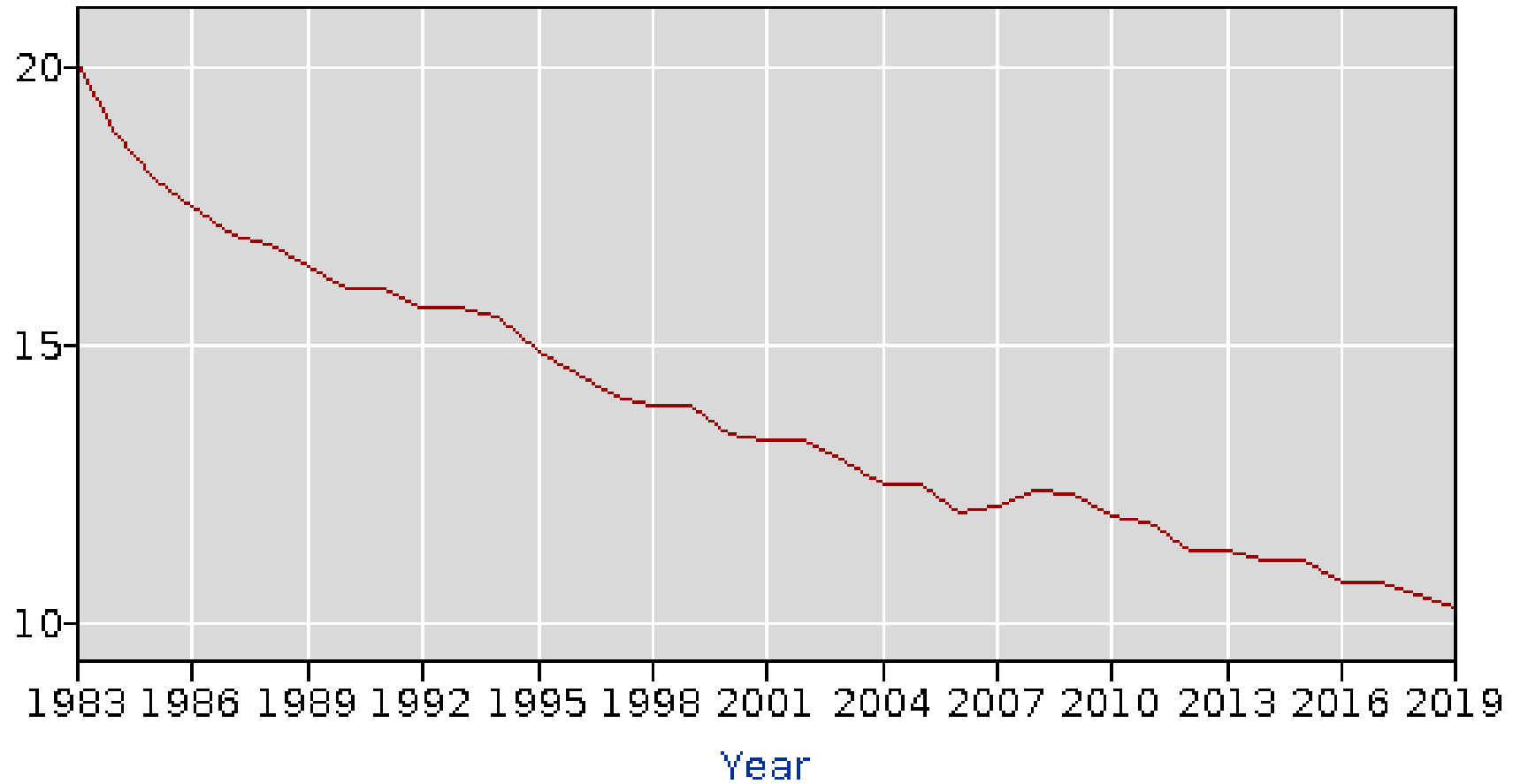
Union Membership (1930 - 1970)

Union Membership as a Percent of the non-Farm Labor Force (1930 - 1970)



Source: United States. Bureau of the Census. (1975). *Historical statistics of the United States, colonial times to 1970* (No. 93). US Department of Commerce, Bureau of the Census, Series D 951

Union Membership 1983-2019



Three Reasons: Unions cont.

- When unions raise the wage above equilibrium, quantity of labor demanded falls and unemployment results.
- “Insiders” – workers who remain employed, are better off
- “Outsiders” – workers who lose their jobs, are worse off
- Some outsiders go to non-unionized labor markets, which increases labor supply and reduces wages in those markets.

Three Reasons: Unions cont.

- Are unions good or bad? Economists disagree.
- **Critics:** Unions are cartels. They raise wages above equilibrium, which causes unemployment and/or depresses wages in non-union labor markets.
- **Advocates:** Unions counter the market power of large firms, make firms more responsive to workers' concerns.

Three Reasons: Efficiency Wages

- The theory of **efficiency wages**: Firms voluntarily pay above-equilibrium wages to boost worker productivity.
- Different versions of efficiency wage theory suggest different reasons why firms pay high wages.

Three Reason: Efficiency Wages cont.

1. Worker health

In less developed countries, poor nutrition is a common problem. Paying higher wages allows workers to eat better, makes them healthier, more productive.

2. Worker turnover

Hiring & training new workers is costly.

Paying high wages gives workers more incentive to stay, reduces turnover.

Three Reason: Efficiency Wages cont.

3. **Worker quality**

Offering higher wages attracts better job applicants, increases quality of the firm's workforce.

4. **Worker effort**

Workers can work hard or shirk. Shirkers are fired if caught. Is being fired a good deterrent?

- Depends on how hard it is to find another job.
- If market wage is above equilibrium wage, there aren't enough jobs to go around, so workers have more incentive to work not shirk.

Activity: Reducing Frictional Unemployment

Which of the following would be most likely to reduce frictional unemployment?

1. The government eliminates the minimum wage.
2. The government increases unemployment insurance benefits.
3. A new law bans labor unions.
4. More workers post their resumes at LinkedIn.com, and more employers use LinkedIn.com to find suitable workers to hire.
5. Sectoral shifts become more frequent.

Activity: Reducing Frictional Unemployment

1. The government eliminates the minimum wage.
3. A new law bans labor unions.

*These are likely to reduce
structural unemployment,
not frictional unemployment.*

Activity: Reducing Frictional Unemployment

2. The government increases unemployment insurance benefits.
5. Sectoral shifts become more frequent.

*These are likely to increase
frictional unemployment, not reduce it.*

Activity: Reducing Frictional Unemployment

4. More workers post their resumes at LinkedIn.com, and more employers use LinkedIn.com to find suitable workers to hire.

*Likely to speed up the process of
matching workers & jobs, which
would
reduce frictional unemployment.*

Key Takeaways

- The unemployment rate measures those actively seeking work but doesn't capture discouraged or underemployed workers.
- Frictional, structural, and cyclical unemployment reflect different causes and durations of joblessness.
- A natural rate of unemployment persists even in healthy economies due to job search and skill mismatches.
- Policies like unemployment insurance, unions, and minimum-wage laws influence unemployment levels and durations.
- Understanding unemployment dynamics helps policymakers design more effective labor-market interventions.

Knowledge Check

1. What is one limitation of the unemployment rate as a measure of labor-market health?
2. How does unemployment insurance affect frictional unemployment?
3. Which type of unemployment arises when workers' skills no longer match available jobs?
4. What happens when unions raise wages above the equilibrium level?
5. Why might firms choose to pay efficiency wages above the market rate?

Knowledge Check Answers

1. It excludes discouraged and underemployed workers.
2. It tends to increase frictional unemployment by reducing urgency in job search.
3. Structural unemployment.
4. It creates a labor surplus—some workers become unemployed.
5. To increase worker productivity, morale, and reduce turnover.